

School Of Computing and Informatics

Systems Analysis & Design

Femmora

Spring – 2026

Contents

1. Feasibility Analysis 2
1.1 Business Overview & Strategic Goals 2
1.2 Proposal 3
1.3 Business Problem 4
1.4 Proposed Solution 4
1.5 Technical Feasibility 5
Availability of Technology 5
Expertise and Skills 5
Compatibility 5
Project Size & Familiarity 5
Overall 6
1.6 Economic Feasibility 6
One Time Costs 6
Annual Costs 7
Benefits 7
ROI & Break-Even Analysis 8
Overall 8
1.7 Legal and Regulatory Feasibility 8
Business Registration 8
Electronic Transaction Law 8
Consumer Protection Law 9
Personal Data Protection Law 9
Overall 9
1.8 Desirability 9
Femmora Desirability 9
Customers Desirability 9
1.9 Viability 10
2. System Analysis 10
2.1 Methodology 10
Selected Methodology 10
Criteria for Evaluating the methodology 11
2.2 The Investigation 11
Observation 11

Questionnaire 12
2.3 Identifying Requirements 17
Functional Requirements 17
Non-Functional Requirements 18
System-level Requirements 18
User-level Requirements 19
Constraints 19
2.4 Analysis Tools 20
Use Case Template 20
Activity Diagrams 21
User Stories & Acceptance Criteria 22
3. System Design 24
3.1 Mockup 24
Customer 25
Admin 29
3.2 Use Case Diagrams 32
3.3 Data Flow Diagrams 33
Context Level 33
Level 0 33
Level 1 34
3.4 Class Diagrams 35
3.5 Sequence Diagrams 36
3.6 Entity Relationship Diagrams 37
References 37

1. Feasibility Analysis

1.1 Business Overview & Strategic Goals

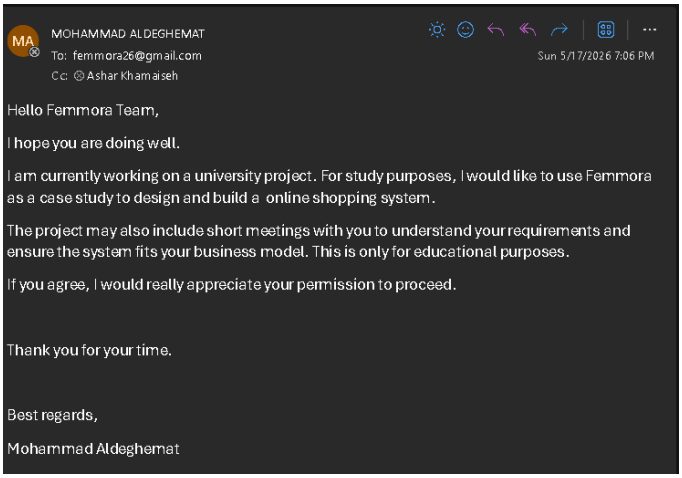
The selected organization is Femmora, a shop founded in 2026 that sells women dresses in Jordan through Instagram. They rely entirely on Instagram posts and stories to present their products, while customers place orders by sending direct messages to Femmora's Instagram account, and the payment process is done through two ways, the first one is CliQ transfer, and the other one is cash on delivery.

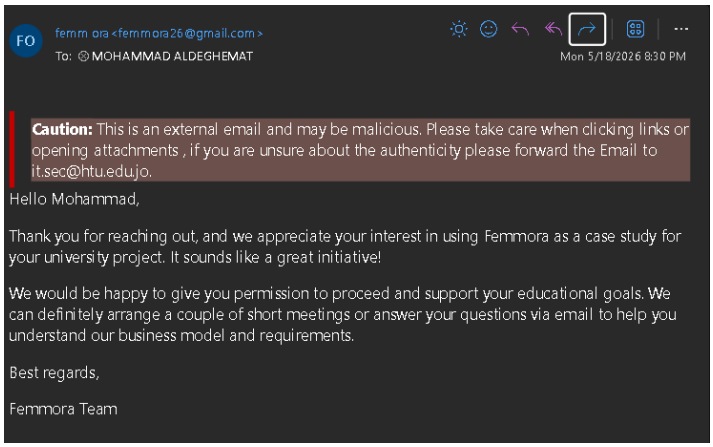
After talking with Femmora's founders' multiple times, I saw that they have specific strategic goals such as:

- Market and Business Growth: Because they are founded in February 2026, they reached a good number of customers but still not the desirable number of customers. Also, on the long-term, they are planning to enter other markets like GULF countries, Syria, Iraq, Palestine, etc.
- Improve Customer Satisfaction: Femmora wants to increase the customer satisfaction by enhancing the customer experience and improve the order process.
- Increase Profit: Femmora always try to increase profit by trying to reach more customers.
- Strengthen Brand: Femmora want to build a strong brand name in the fashion market.
- Always Improve Service Quality: Femmora always try to improve service quality in terms of clothes quality, and customer service quality.
- Automate process: Femmora wants to automate the overall process in terms of placing the order, and inventory management process.
- Memorize loyal customers: Femmora wants to memorize loyal customers, but it may be hard without a structured customer database.

1.2 Proposal

I sent an email for Femmora at 17th of May, they responded with an acceptance at 18th of May, and they were open for the idea and for meetings.





1.3 Business Problem

The current manual process creates a lot of problems that has a direct effect on operational efficiency, customer experience and satisfaction, and the growth of the business. The main problems are:

- Poor Product Management Process: Products are sometimes updated which is difficult to do in an Instagram Page.
- Poor Order Management Process: Orders are managed through Instagram direct messages which makes it difficult to track, organize, or retrieve order information effectively.
- No Order Tracking for Customers: Customers cannot track their order status unless they message the shop to get updates on their order.
- No Customer Data Management: There are no saved records of who ordered before which makes it difficult to know who the loyal customers are and even view their information.
- Poor Inventory Management: When order is received, the owner must check the physical inventory to see if the product is in stock or out of stock.
- No Sales Analytics: Without data saved in a database, the owner cannot identify best-selling items, and profit status or performance.
- Current System Cannot Support Business Growth: When the orders increase and become a lot, the current process cannot handle a large number of orders effectively.

1.4 Proposed Solution

Based on the problems we discussed, the proposed solution is to convert the whole process into an automated process through building a web-based application that serves the e-commerce and order management processes, solve the current system issues, and aligns well with the strategic goals for Femmora.

The system will connect customers and staff in one platform that eliminate the current reliance on direct messaging, and manual order processing by introducing new ways to handle the overall process.

The system will improve the workflow by making sure that the staff can manage products, orders, inventory, and customer information easily, and view a detailed dashboard of sales performance which helps the staff to control the operations in an efficient way, reduce human error, and access information easily.

The system will provide a single database for all orders, customers, and products which helps the staff to track and access information effectively. It will also reduce human error and making sure that the data are consistent.

The system will support a strategic goal for the business which is business growth by making the process handle increasing numbers of customers and transaction without lowering the service quality.

The system will provide a better experience for the customers by making the customer interact with the business through a system instead of direct messaging which

improves communication efficiency, reduce delays, and increase customer satisfaction.

Also, the system will provide real-time order updates for the customer which makes it easier for the customer to track their orders.

Overall, the solution will improve all of the business operations and take Femmora to a next level of professionalism.

1.5 Technical Feasibility

Availability of Technology

All of the required technologies used to build a web-based e-commerce system are widely available, and well-documented. The system can be developed using modern technologies such as React.js or Next.js for the front-end, Node.js or PHP for the back end, and MySQL for the database management system, and these technologies are widely available and used by a lot of developers and are well-documented.

Cloud services are also widely available in the region and offer plans that are suitable for small businesses like Femmora such as AWS and Google Cloud. Payment Gateways are widely available and can be easily integrated with the system without any customization.

Expertise and Skills

The development of the system requires experts in web development, database design, and UI/UX design, and there are a lot of talented experts across Jordan and the world. Freelance developers and small software development teams can build an e-commerce system efficiently.

E-commerce systems are a well-understood area of software, and the development team works with familiar technologies and does not work with unfamiliar or emerging technologies which means that developers does not need to invent solution, but they can follow best practices and implement the features desired by Femmora.

Compatibility

Since Femmora currently does not use existing IT systems, there isn't any concerns related to compatibility with legacy infrastructure. The system will be built from scratch, with no system to modify or replace and no integration requirements except integration with the payment gateway which eliminates any risks associated with integration.

The system will have a responsive design to make sure that the system will be compatible with different browsers such as chrome and edge and can be used on different devices whether it's a desktop or mobile which make sure that all the customers can access the website from anywhere. Also, this responsive design supports the strategic goal of market growth because as technologies improve, a new future screen sizes and device types will be introduced and Femmora can be accessible on these new devices.

Project Size & Familiarity

The scope of the system is medium, as it includes products, order management, inventory management, customer accounts, payment, etc. So the estimated time for

this project is four to five months for a team of two or three developers which is very realistic for a small business and a medium system.

Overall

Everything we need technically:

- Cloud Server: To host the website and the data base we need a cloud server such as (AWS)
- Front-end technologies: HTML, CSS, React, Next
- Back-end technologies: PHP, Django, Node
- Database technologies: MySQL
- Open-Source IDE's: VS CODE, Eclipse, etc.
- Payment gateways: PayPal, Visa, MasterCard
- Laptops.
- Developers.
- Designers.
- Testers.
- Marketing team.
- Project Manager.

So we can say that the system is technically feasible because the required technologies are available and well-supported. The development, design, and testing expertise are available and accessible at a cost within the project budget. The system does not need to be compatible with legacy infrastructure because there isn't any legacy system that the new system should be integrated with. There are no unknown technical areas that would prevent the successful development of the proposed system.

1.6 Economic Feasibility

Since the Business in newly founded, everything in this analysis was based on a reasonable assumption.

One Time Costs

Item	Cost
UI/UX Design	$500
Web Development	$3,000
Testing & QA	$650
Buy a Domain	$20
Cloud Hosting Setup & Configuration	$100
SSL Certificate	$30
Payment Gateway Integration & Setup	$250
Total Costs	**$4,550**

I did not include any separate labour costs because the proposed system is small and Femmora budget is relatively small, so, instead of hiring full-time employees, we will

hire freelancers that their costs is already included within the design, development, and testing expenses. Also, the estimated costs for all of these activities were intentionally set higher than the expected actual costs to have a reserve cost for any unexpected costs that may arise during working on the project.

Annual Costs

Item	Cost
Cloud Hosting	$300
Domain Renewal	$20
SSL Certificate Renwal	$30
System Maintenance & Support	$200
Payment Gateway Fees	$200
Total Costs	**$640**

Benefits

Intangible Benefits:

- Enhance Femmora's brand image and appear as a professional online shop.
- Improve customer satisfaction by providing faster and smoother shopping experience.
- Improve customer trust and loyalty through providing them a reliable shopping experience.
- Better decisions are performed through sales and performance dashboard and analytics.
- Compete more in the market by allowing Femmora to operate as a professional brand.
- Reduce confusion and human errors by viewing orders, inventory, customers, and products in one system and database.

Tangible Benefits:

Item	Benefit Value	Assumptions
Increased Sales Revenue	$1500	Grow 40% per year as assumed based on femmora's owner expectations.
Labour Cost Saving on manual order processing	$400	Femmora owners said they pay for someone to manage to the messages and place orders
Reduction in order errors	$150	Femmora owners said they face order errors 3 times a month approximately
Total Benefits	**$2,050**	

ROI & Break-Even Analysis

	Year 0	Year 1	Year 2	Year 3	Year 4
Total Benefits	$0	$2,050	$2650	$3,490	$4,666
Total Costs	$4,550	$640	$640	$640	$640
Net Benefits	$0	$1,410	$2,010	$2,850	$4,026
Cumulative Net Cash Flow	-$4,550	-$3,140	-$1,130	$1,720	$5,746

ROI = (Total benefits – Total cost) / Total cost

= (12856 - 7110) / 7110

= 5,746 / 7110

= 0.81

ROI = 81%

Break-Even: as shown in the table above, the break-even would be in between year 2 and year 3 to become positive.

BEP = n + cumulative cash flow at year n / net cash flow at year n + 1

= 2 + 1130 / 2850

= 2 + 0.39

BEP = 2.39 years

Overall

Based on the economic analysis and based on cost vs benefits, the system worth the investment. The total cost is reasonable for a system in that scope and it was verified through current freelance costs. The ROI of 81% is a very strong result which means that for each $1 invested, Femmora get 1.81 in return over the four years. The break-even point of 2.39 years is very strong for a startup, especially since 2026 and it still building their customer base.

1.7 Legal and Regulatory Feasibility

All organisations must follow all the legal and regulatory aspects of its industry, and in this part, we will review different legal and regulatory aspects for Femmora system.

Business Registration

To operate legally in Jordan as an e-commerce system, Femmora must register its business as a legal business entity. The registration is obtained through the Ministry of Industry and Trade.

Electronic Transaction Law

Jordan’s Electronic Transaction Law provide the framework for e-commerce, digital signatures, and online contracts to legally operate. Femmora system must comply with the Electronic Transaction Law through providing clear terms and conditions for purchases, make sure that the contracts are legally valid, and save the transaction

records. The Femmora system will provide clear terms and conditions for purchases and make sure that contracts are performed legally.

Consumer Protection Law

Jordan's Consumer Protection Law provide the obligation for businesses that sells goods for consumers such as clear disclosure for prices, return and refund policies, and product description. The Femmora system will display product description, prices, and return and refund policies will be stated on the platform to make sure that the system is complied with the consumer protection law.

Personal Data Protection Law

Jordan's Personal Data Protection Law governs the collection, storage, and processing of personal data of Jordanians. The Femmora system will collect only the minimum necessary data, store customer data securely and encrypt them, provide the customers with a privacy policy, allow customer to delete their data if they wany, and make sure that none of the customer data is sold or shared to third parties.

Overall

Overall, the Femmora system is legally feasible only if it will be built while based on the compatibility with these Jordanian Laws such as the Electronic Transaction Law, Consumer Protection Law, and Personal Data Protection Law. None of these laws require complex technologies. Failure to comply with these legal requirements may result in penalties, loss of customer trust, data loss, or even facing prison time.

1.8 Desirability

Femmora Desirability

The owners clearly indicated that there are a lot of limitations in their current Instagram based operation. The manual processing of managing orders through direct messages, manually track the inventory, and find order history is time-consuming and there is a high possibility of errors. The owners discussed the strategic goals that align with the desired system such as automate the process, market growth, increase profit, improve brand name and customer satisfaction, and memorize loyal customers, and these strategic goals have a clear connection with Femmora owners' desire.

These goals and desires are maintained through the system, as the system will support automating the process, market growth, increase profit, and improve brand name through more visibility and sales analytics dashboard that guides them through decision making, improve customer satisfaction through providing them an easier way to place their orders, memorize loyal customers through a structured centralized database that save records for everything included in the process, and all of that cannot be achieved with the current process but can be achieved with a dedicated system.

Customers Desirability

The current Instagram based process lack the transparency that customers expect, as most of the e-shops operates through a dedicated system and not through a social media platform. In the current process, customers are required to send direct messages to place their order, wait for response to confirm availability, arrange payment, arrange delivery, and then always ask for order updates. This process come with lack

of clarity, delays, low customer trust and satisfaction, and uncertainty in the customer journey.

The system will eliminate all of these points and provide a self-service experience as customer can prowse products, view availability, add items to the cart, choose payment method, receive order confirmation, and track their order status without the need of talking to Femmora through social media which eliminate delays, enhance clarity, customer trust, and satisfaction. These criteria are now considered the basic standards for customers to place online orders and Femmora currently have a desirable gap for customers, and the proposed system will fill this gap.

1.9 Viability

Femmora is a small business startup that does not have any existing IT infrastructure or large legacy systems to be integrated with or to replace, which lowers the risk of integration and adoption. Since the system is built from scratch, and the Owners are already active, their only technical risk is not knowing how to use the web-based platform, but after contacting them, they stated that they know how to use a web-based platform and does not need any training.

Also, to assess the viability, we need to assess the market. The e-commerce market in Jordan and middle east is experiencing a growth because of expanding digital payments and increasing the people who shop online. The women's fashion segment is one of the categories that perform well in the e-commerce industry. So, the transition from Instagram to a dedicated platform would make Femmora capture a large share of the growing market. Also, the system scalable architecture makes sure that the system can handle large users and transactions as the business expands to Gulf and neighboring countries which makes the sustainability of a long-term solution strong.

Another thing is that the cost of $4,450 is reasonable for this stage, and with ROI of 81% and a break-even of 2.39 years justify why Femmora should invest in that system. There are low indicators of financial risks that would threaten the system's long-term sustainability.

2. System Analysis

2.1 Methodology

Selected Methodology

The selected methodology for developing Femmora system is Scrum which is categorized in the Agile development framework.

Scrum is an iterative approach that focuses on the software, customer collaboration, and adapting with requirements change which allow Femmora owners to review the progress, provide feedback, and collaborate with us with requirements change.

Scrum was selected due to many reasons such as the nature of Femmora project, as it needs flexibility, continuous stakeholder involvement, iterative development, and the need to adapt for new changes in business requirements. Femmora is a small business

and still considered a startup that compete in an environment that frequently changes, so it is expected that some businesses requirements may change or introduced during the development of the Femmora system.

The iterative approach of scrum would allow Femmora to review and assess important features and identify requirements change or problems earlier which reduces the risks and minimize the costs since Femmora does not have a very big capital.

Also, Scrum requires a very high customer involvement, and since the owners are available most of the time, there will be no delays waiting to talk with the owners which means that Femmora owners are always involved in the process and can participate through the development process and continuously provide feedback which allow Femmora and us build the system to reach the strategic goals and requirements and deliver a high-quality product.

User satisfaction and user experience play a critical role in e-commerce system, and with the continuous validation and owners feedback we can achieve and improve the overall user satisfaction UI/UX, and user journey.

Also, with scrum, we can prioritize features based on business value and customer needs, and based on Femmora capital, we can only introduce the most important features first and add less important feature later.

For Femmora, we are hiring a team of freelancers, and scrum support that by having A cross-functional teams that have a diverse of skills which reduces the costs of having to hire a lot of employees that are specialized in one field.

Criteria for Evaluating the methodology

1. Requirement Stability
2. Stakeholder Involvement
3. Adaptability to change
4. Cost
5. Risk of failure

2.2 The Investigation

Observation

I conducted Participant observation by working within Femmora's operations as a delivery driver and participating in the order preparation process which gave me a better understanding of the real workflow and environment that could not be fully understood through interviews or surveys. The process of observation wasn't difficult, as the business belongs to people I personally know so I was more than welcomed to participate in the business operations.

Observation was chosen because it provides a real world understanding of how the business actually operates instead of relying on what owners describe in interviews, as people might not fully describe problems or gaps, but observation helps me see hidden problems and gaps and fully understand them. Observation make sure that the proposed system is built around the real workflow and not just a description of it.

However, Observation has some limitations as it can be time-consuming because understanding the workflow requires participating for a period of time.

Observation support requirement gathering through:

- It provides a practical understanding of how the business work in the real world.
- It helps me identify daily problems that Femmora faces.
- It helps me to understand how orders are prepared and managed from placing the order till the arrival.
- It helps me to understand how inventory is managed.

After participating in the business process and communicating with Femmora's owners and team members, **I concluded that:**

- Order Preparation Process: Orders are prepared manually by checking Instagram messages to see what each customer ordered and there wasn't any sheet or reference which means the person who is preparing the order must return to the messages to verify order details.
- Inventory Management: Managing the inventory was time-consuming, as they have to check the storage manually when customers orders and there weren't any centralized records which means if a customer ordered something and it was placed without checking the inventory, then the order must be cancelled.
- Order Status: Customers always ask for the order status, and tracking which orders are delivered, pending, have issues were handled through repeatedly update the customer through direct messages.
- Peak Period Pressure: During high order period such as Eid, the manual process became inefficient, and they needed to hire more people to reply to direct messages so that they handle and confirm all the orders which also take so much time in inventory and stock verifications.

Questionnaire

I also conducted a Questionnaire to understand the requirements of the customers side of the system. The survey was distributed to people who usually order online and especially through Instagram shops to understand their experience with Instagram order process and identify what these customers expect from an e-commerce platform. I collected 31 responses from different users that usually order from Instagram pages.

Questionnaire was chosen because it provides structured data and a large number of insights that help us to collect multiple opinions and insights with short period of time and low cost, as I needed a good number of insights, so I provide accurate and suitable requirements. In comparison, interviews would make us collect a very low number of insights and people may feel less comfortable expressing opinions in interviews. Also, because there are a good number of responses, questionnaires make it easier to identify patterns and customer expectations. Questionnaire was conducted only to the customer side because customer base is large and their opinions vary. In contrast, the Femmora team is small, so observation would be more suitable technique to understand the Femmora side and their operations.

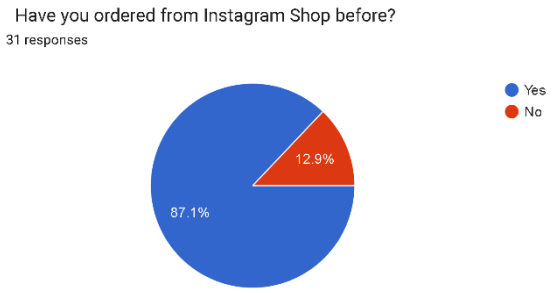
However, Questionnaire has some limitations as some people may fill the questionnaire randomly which gives us inaccurate data.

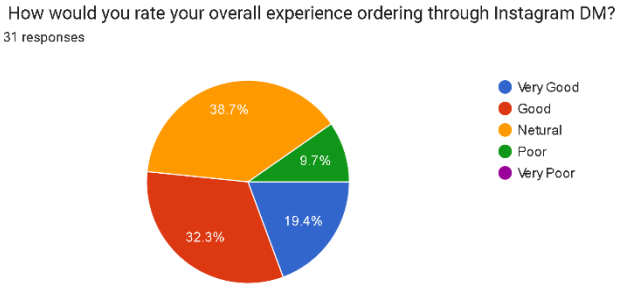
Questionnaire supports the requirement gathering process through:

- Validate assumptions that were identified during observation.
- Collect a large number of data to make an accurate requirement determination process.
- Identify which features are the most important for the users.
- The mix between open-ended and closed-ended questions provides us with detailed responses and is useful for gathering specific information.

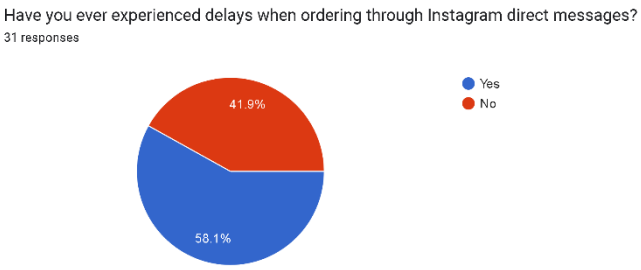
Questionnaire link: [Build Femmora With Us - Google Forms]

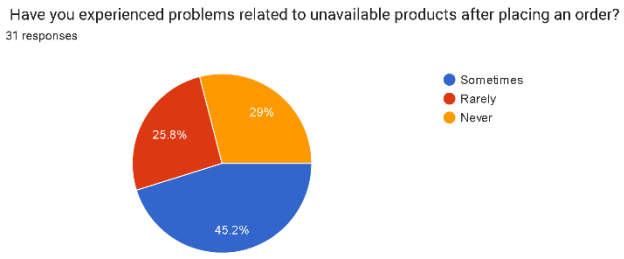
Findings





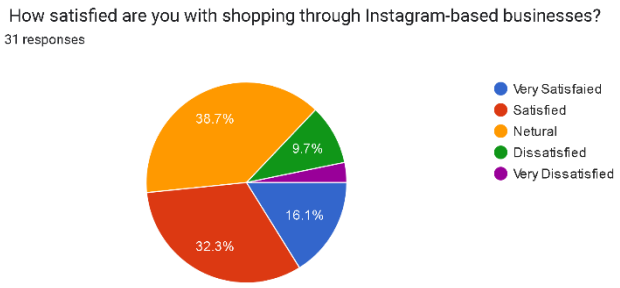
87.1% of respondents said they previously ordered from Instagram shops which shows that the responses are based on real experiences. But the overall experience of shopping through Instagram direct message was not highly positive, as most of the respondents rated their experience as good of neutral while some users rated their experience as poor or very good which shows that there are some problems that affect the customer satisfaction and we want to improve that experience for the customers of Femmora.

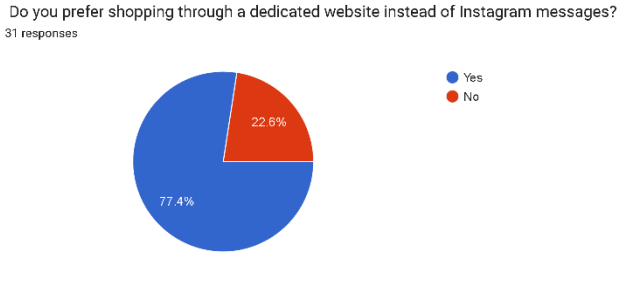




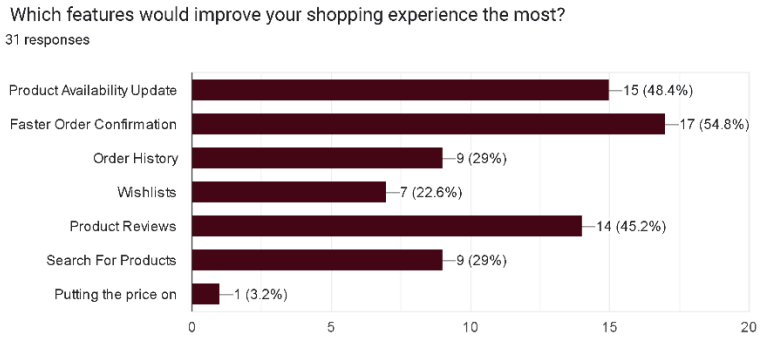
58.1% of the respondents experienced delays when they order through Instagram page which shows that the new system must support faster order management and support. Also, 45.2% said the sometimes experience problems related to unavailable products

after placing the order which means that the new system must support inventory management system with stock updates.





Most of the customers were satisfied with shopping through Instagram but they prefer shopping through a dedicated website instead of Instagram messages as 77.4% prefer the website and 22.6% prefer shopping through Instagram which shows a strong customer acceptance of the new system.

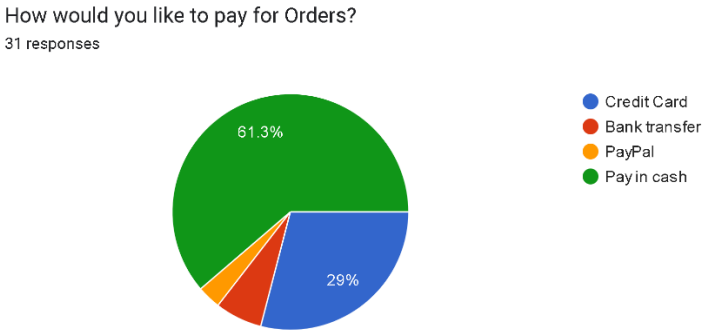


When I asked about the features that would improve their shopping experience some features were frequently selected and some are not.

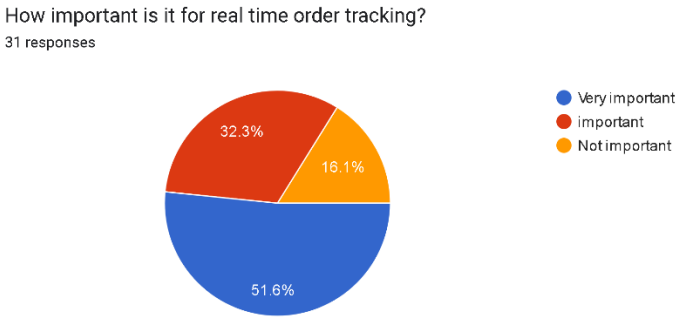
Features that more than 45% of the respondents selected:

- Product Availability
- Faster Order Confirmation
- Product Reviews

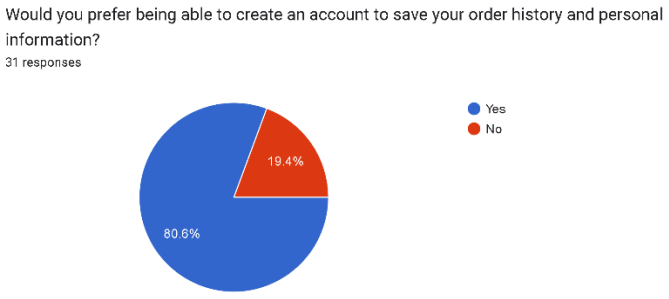
Which means these are the most important features that would improve customer shopping experience, while the other features are also essential such as order history and product search.



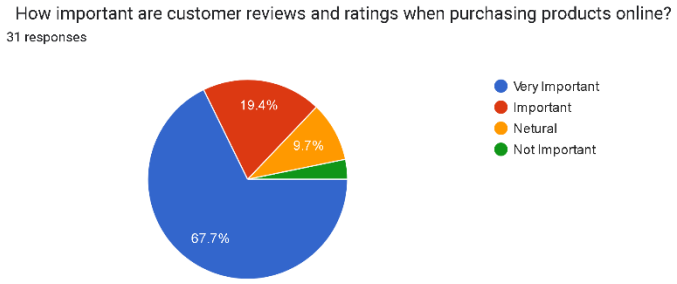
Most of the respondents preferred to pay in cash, but some smaller percentage preferred online payment methods such as credit cart which means that there should be different options for payment to satisfy customer preferences.



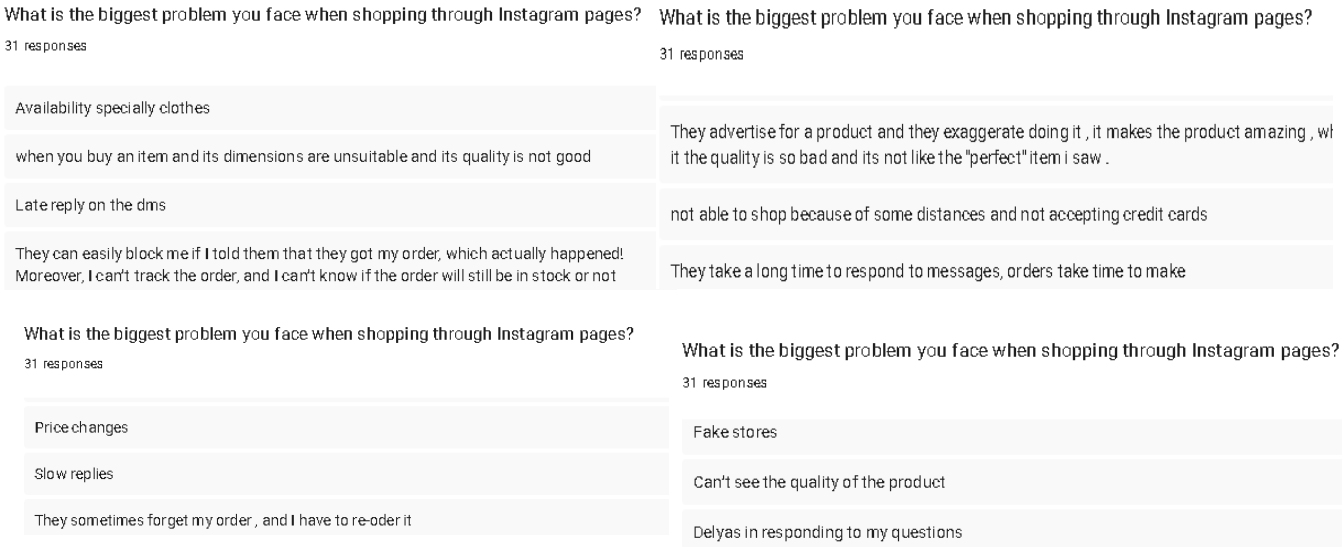
51.6% of the respondents said that real time order tracking is very important, and 32.3% said it is important, which means that more than 80% of respondents consider order tracking as an important feature within the system and it would raise customer satisfaction.



80.6% said that they prefer to be able to create accounts, save their order history and personal information which would also help Femmora owners to view all order history, and customers information.



As stated earlier, customer reviews and rating is one of the important features, and 67.7% of the respondents said that it is very important for the system to have that feature.



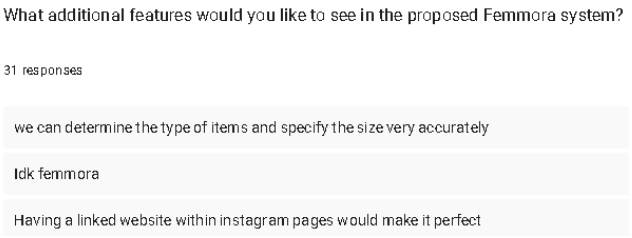
This open-ended question revealed problems that we did not think of before, and one of the most common issues mentioned by respondents was slow communication and delay in replying to customer messages which shows that relying entirely on Instagram direct messages create problems in communication that affect the customer satisfaction and trust, so the system must include faster order confirmation, and centralized order management to eliminate this issue and raise customer satisfaction.

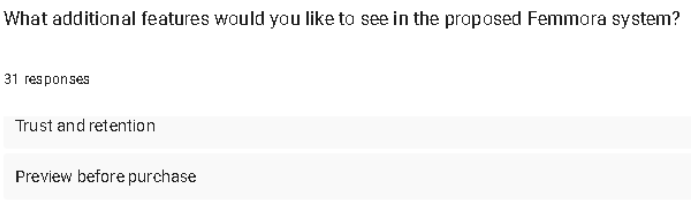
Also, respondents said that one of the problems is availability, so when customers are excited to place an order, they must communicate with the shop, then find out it's not available they get frustrated, so the system must only show the items that are available in stock.

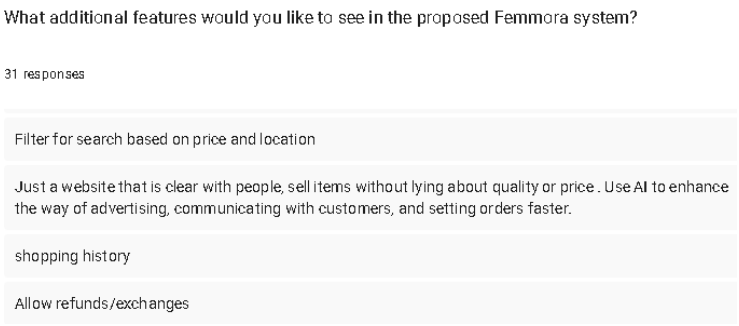
Also, respondents said that product size might vary and might not be the best way to measure the size, so there must be more details on the product size such as height and width.

Other problems like payment methods and customers' fear from fake stores will be resolved through a dedicated website that is acknowledged by the Ministry of Industry and Trade and with multiple payment methods.

Other responses were not useful, as they answer it with a dot or something irrelevant.







The other open-ended question did not provide a lot of useful insights but it provided some valuable insights and what the customers expects. Respondents said they want trust, transperancy, and product accuracy which support the detailed product pages, accurate size information, multiple product images, and customer reviews.

Also, they requested the search and filter functionality based on price, size, etc. Which shows that customers expect an easy product navigation and discevory and the new system should include the search and filtering functionality.

Femmora will still operate through Instagram for questions and advertisements, but placing order process is done via website as stated in one of the responses.

2.3 Identifying Requirements

Functional Requirements

Functional Requirements describe what the software should do and the specific features the system must provide to satisfy user needs.

- Customer Management:
  - The system shall allow customers to create account and enter their details.
  - The system shall allow customers to view a page with all the products in the shop.
  - The system shall allow customers to easily search for products based on filters such as price and size.
  - The system shall allow customers to add products to the cart.

    - The system shall allow customers to customers view products in their cart.
    - The system shall display detailed product information and stock availability.
    - The system shall allow customers to place orders without even contacting the shop.
    - The system shall allow customer to track their orders through (preparing, on delivery, delivered) states.
    - The system shall allow customers to review and rate products.
    - The system shall allow customers to pay in different payment methods including online payment and cash on delivery.
- Staff/Admin Management
    - The system shall allow administrator to add, update, and remove products.
    - The system shall allow administrator to view all orders in the system.
    - The system shall allow administrator to manage stock when a shipment arrives through increasing the number of a specific product.
    - The system shall allow administrator to update order status for customers so that customers track their orders.
    - The system shall allow administrator to view all customers in the system.
- Reporting
    - The system shall allow staff to view a monthly summary sales report including total sales, and top-selling items.

Non-Functional Requirements

Non-functional requirements are the characteristics the system should have such as performance, security, reliability, scalability, and usability.

- The system shall restrict unauthorized access.
- The system shall have clear permissions for users.
- The system shall use data encryption techniques on passwords and payment data.
- Thes system shall be developed using an architecture that accept future updates.
- The system shall be accessible on different browsers.
- The system shall be responsive as users can open it on different screen sizes.
- The system shall have 99.9% uptime.
- The system shall load pages withing approximately 4 seconds.
- The system shall handle 5000 active users at the same time.

System-level Requirements

System Requirements describe how the system should be built such as how it should be designed, implemented, and operated withing its environment.

- The system shall use a centralized database to store customer, product, inventory, and order information.

- The system shall operate as a web-based application accessible through different browsers.
- The system shall integrate with different payment gateways to support different payment methods and process transactions securely.
- The system shall implement access based on roles.
- The system shall automatically update inventory records when an order status is preparing.
- The system shall backup data regularly to avoid data loss.

User-level Requirements

User Requirements describe what the users need to do as it explains the goals of users and what do they need to achieve.

1. The customer shall be able to create user account and enter their details.
2. The customer shall be able to manage their account and access their order history and their information independently.
3. The customer shall be able to view all products in the shop.
4. The customer shall be able to search for products and add them to cart without the need to contact the shop.
5. The customer shall be able to view all products in their cart including quantity and price.
6. The customer shall be able complete the full process from selecting a product to order confirmation without the need to contact the shop.
7. The customer shall be able to track their order status without the need to contact the shop.
8. The customer shall be able to choose their preferred payment method based on their preference.
9. The customer shall be able to share and view product reviews and ratings.
10. The customer shall be able to view product details such as size, dimensions, stock availability and product images.
11. The admin shall be able to have a full control managing on the products in terms of adding, updating, and deleting products.
12. The admin shall be able to view all orders in the system.
13. The admin shall be able to update inventory records.
14. The admin shall be able to manage order status and mark them as preparing, delivering, delivered.
15. The admin shall be able to monitor business performance through reports related to sales including sales performance and best-selling item.
16. The admin shall be able to view all customers in the system.

Constraints

There are some constraints specified as organizational and regulatory constraints that will be listed below.

- Organizational Constraints:
    - Femmora has limited capital which could limit the number of features that can be developed in the first release.
- Regulatory Constraints:

- As mentioned in the legal feasibility there are laws that Femmora must comply with in order to build some of the requirements such as Electronic Transaction Law, Consumer Protection Law, and Personal Data Protection Law.

2.4 Analysis Tools

Use Case Template

Use case name: Place an Order	**ID:** UC - 1	**Priority**: Very High
Actor: Customer		
Description: This use case describes the process when a customer browse product, select products, add product to cart, and place their order		
Trigger: Customer clicks the "Confirm Order" button on the platform. **Type: External**		
Preconditions: • Customer created an account • Customer login to the system with his account • Customer added a product to the cart • System validates if product in stock		
Normal Course: • Customer browse products on the system • Customer add one or more products to the cart • Customer open cart and view order summary • Customer enter details such as their full name, delivery address, and phone number • Customer Specify payment as online payment through credit card • Customer enters credit card details through payment gateway • System validates the entered details • System display order confirmation page with order ID		
Alternative Course: • Customer select another online payment method such as PayPal • Customer select another payment method such as pay in cash which will eliminate entering the online payment details • Customer remove a product or update product quantity before confirming the order • Customer save the cart and complete the ordering process later		

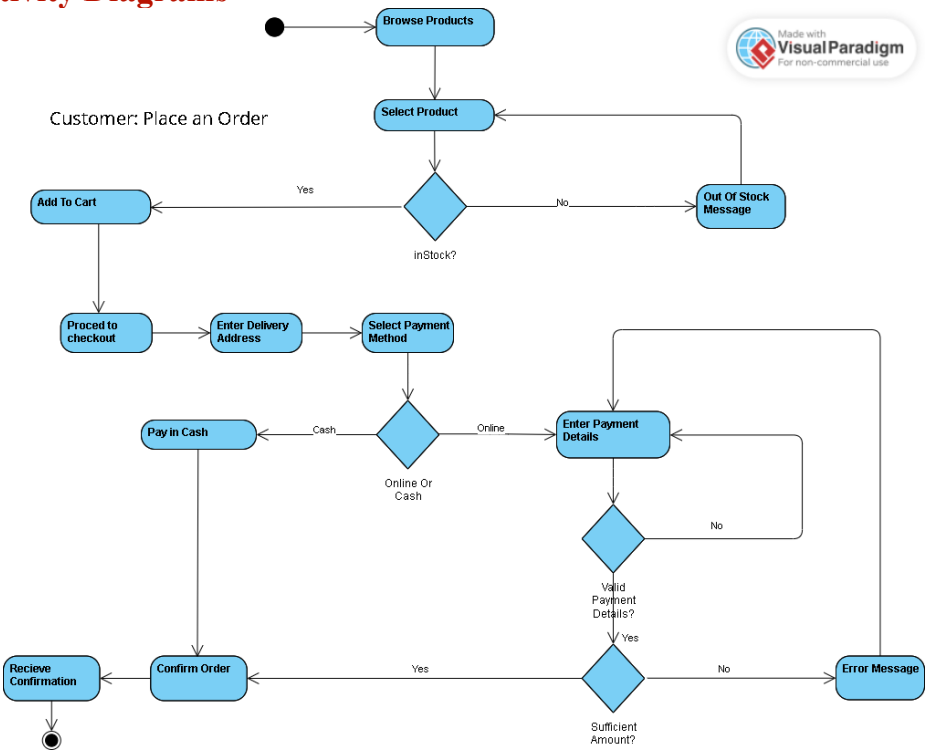
Postconditions:

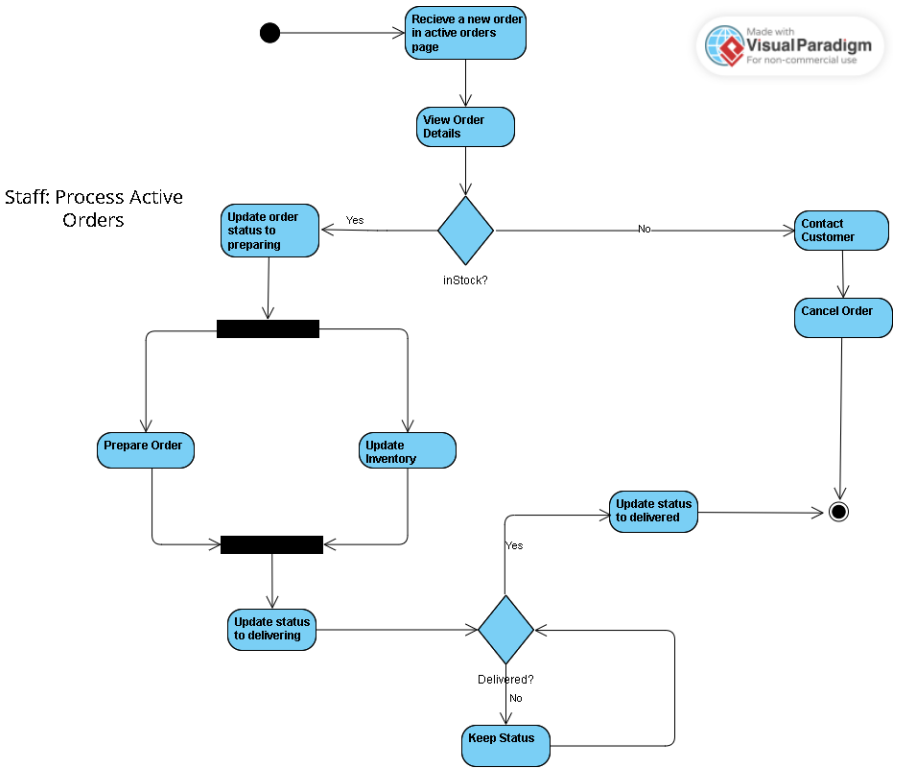
- A new order record is stored in database with placed status
- Order is displayed in order history for the customer
- Order is displayed in the active orders page for the staff
- Inventory is updated based on the products bought
- Customer's old cart is cleared

Exceptions:

- Product is out of stock that would prevent order confirmation
- Customer leaves required fields empty such as full name, delivery address, and phone number
- Online payment fails due to insufficient amount or not supported card for payment gateway
- Customer searching and try to add to cart without logging in
- The quantity of a product exceeds the available quantity

Activity Diagrams





User Stories & Acceptance Criteria

1. As a customer, I want to create and manage my account, so that I can save my information and access my order history.
    - Acceptance Criteria:
        i. Customer should be able to enter required information to create an account.
        ii. If created successfully system store customer information and allow customer to login.
        iii. Once logged in, customer should be able manage their information and access order history.
2. As a customer, I want to search and browse for products, so that I can find products easily and add them to cart.
    - Acceptance Criteria:
        i. If products are not deleted, the system should display products with their image, prices, and description.
        ii. The customer should be able to search for products using filters, if matching products exists, system should display them.
        iii. If products are available, customer should be able to add products to the cart.
3. As a customer, I want to place orders on the website, so that I can complete a purchase independently.
    - Acceptance Criteria:
        i. Customer should be able to proceed to checkout, if there aren't products in cart, system should not allow customer to complete the process.

        ii. Once proceed, customer should be able to enter information and select payment method.
        iii. If products still available and payment is successful, system should create order ID and display confirmation.
        iv. If payment fails or product become unavailable, system should prevent order confirmation.
4. As a customer, I want to view my shopping cart, so that I validate the products in the cart before placing an order.
    - Acceptance Criteria:
        i. Customer should be able to access shopping cart and view products in it.
        ii. If there are no products in the cart, the system should display "empty cart"
5. As a customer, I want to track my order status, so that I can know the progress of my order.
    - Acceptance Criteria:
        i. Customer should be able to access their active orders, and view order status.
        ii. If staff update the status, then it should be updated on the customer side too.
6. As a customer, I want multiple payment method, so that I can choose my preferred payment method option.
    - Acceptance Criteria
        i. The customer should be able to choose between cash on delivery and online payment options.
        ii. If online payment is selected, system should request for payment details.
        iii. If details are not valid, system should display error messages and prevent order confirmation.
7. As a customer, I want to review and rate products, so that I can view customer's experiences and share mine.
    - Acceptance Criteria
        i. Customer should be able to submit ratings and review for products.
        ii. If review is submitted successfully, system should display it in the product page.
        iii. Then customers can view ratings and reviews submitted by other customers.
8. As a customer, I want to view product details, so that I can decide whether to buy it or not
    - Acceptance Criteria
        i. Customer should be able to open product page, and system should display image, size, description, size, stock.
9. As an admin, I want to manage products, so that I can keep product information updated.
    - Acceptance Criteria

        i. Admin should be able to add/update products, if all required details are filled, system should display the product to the customer.
        ii. Admin should be able to remove products, if removed, system should no longer display the product to the customer.
10. As an admin, I want to view all orders, so that I can check old orders or verify the details of active orders.
    - Acceptance Criteria
        i. Admin should be able to view all orders in the system if there are any.
        ii. Then, if there are orders, system will display customer and order details related to that order.
11. As an admin, I want to manage inventory records, so that I keep the stock availability accurate.
    - Acceptance Criteria
        i. Admin should be able to increase or decrease product quantities in inventory if that product exists.
        ii. Once increased or decreased, it will be updated automatically on that specific product.
12. As an admin, I want to update order status, so that customers keep updated with their order progress.
    - Acceptance Criteria
        i. Admin should be able to enter active orders and update its status.
        ii. Once updated on the admin side, then it should be updated on the customer side too.
13. As an admin, I want to monitor sales reports, so that I can assess business performance and make informed decisions.
    - Acceptance Criteria
        i. System should display total sales and performance for admin, if there were orders in that period.
        ii. If there is no data, system will show an error message.
14. As an admin, I want to view all customers in the system, so that I can access customer information when needed.
        i. Admin should be able to open customer page.
        ii. System should display customers with their order counts if there are any customers in the system.

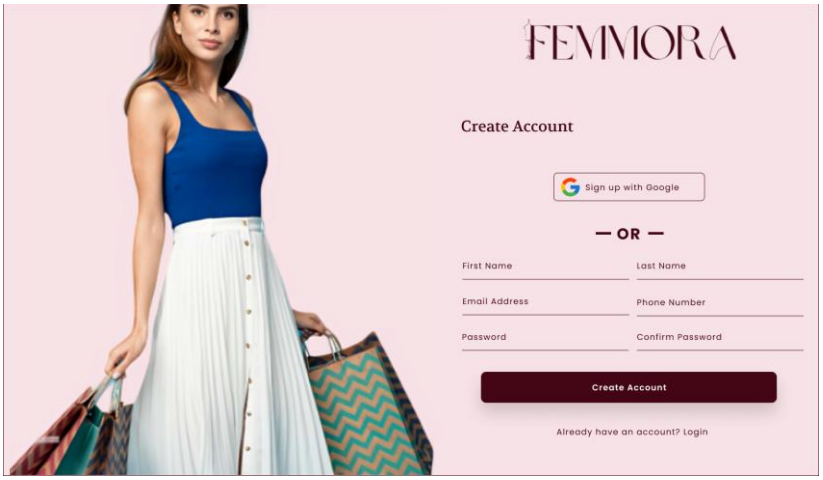
3. System Design

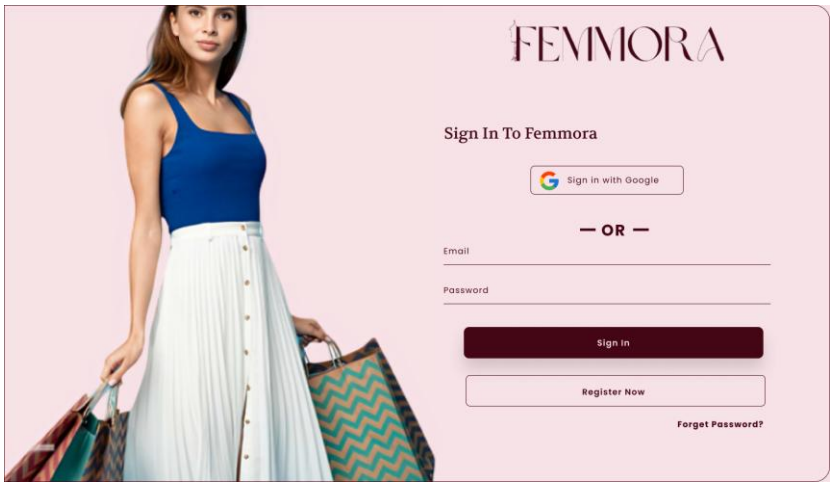
3.1 Mockup

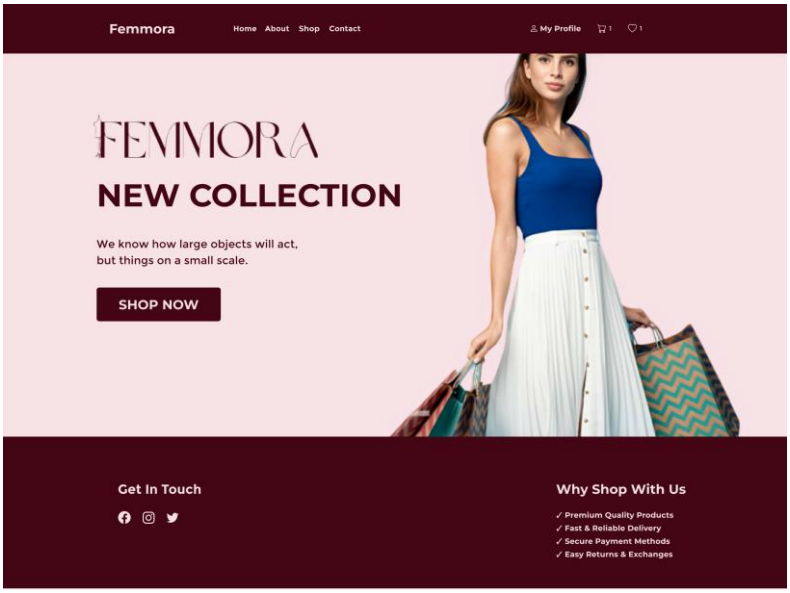
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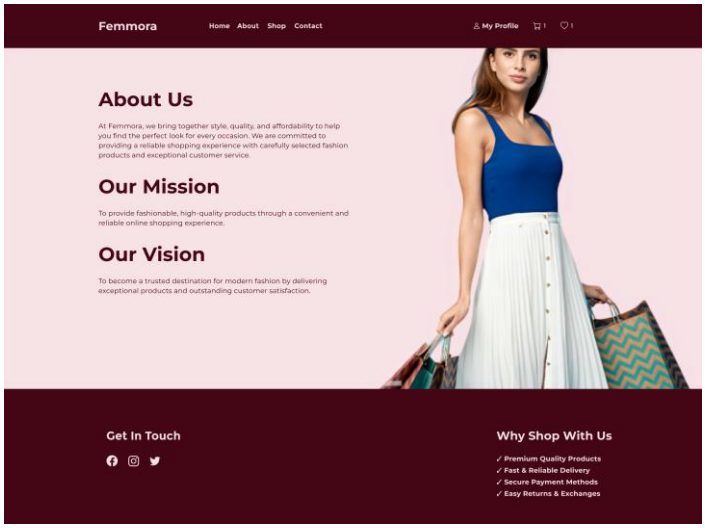
Link: Femmora MockUp

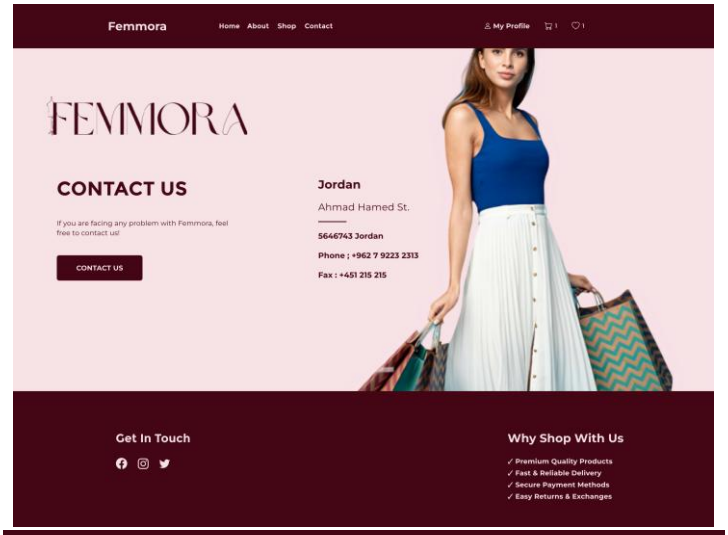
Customer


FEMMORA
Create Account
Sign up with Google
— OR —
First Name
Last Name
Email Address
Phone Number
Password
Confirm Password
Create Account
Already have an account? Login


FEMMORA
Sign In To Femmora
Sign in with Google
— OR —
Email
Password
Sign in
Register Now
Forget Password?


Femmora
Home About Shop Contact
My Profile
FEMMORA
NEW COLLECTION
We know how large objects will act,
but things on a small scale.
SHOP NOW
Get In Touch
Why Shop With Us
✓ Premium Quality Products
✓ Fast & Reliable Delivery
✓ Secure Payment Methods
✓ Easy Returns & Exchanges







MY PROFILE

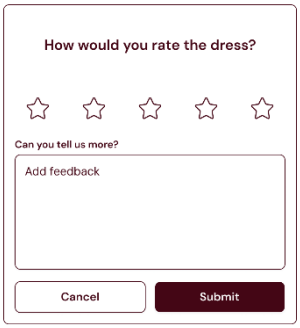
Maya | Mahmoud

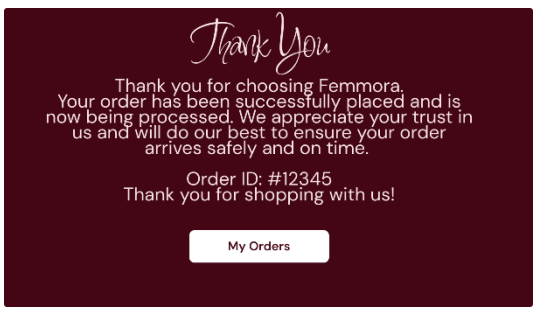
Maya@gmail.com | +962 7 3452 2525

Amman, Jordan, Alnoyayry St., Building Number 1

MY ORDERS

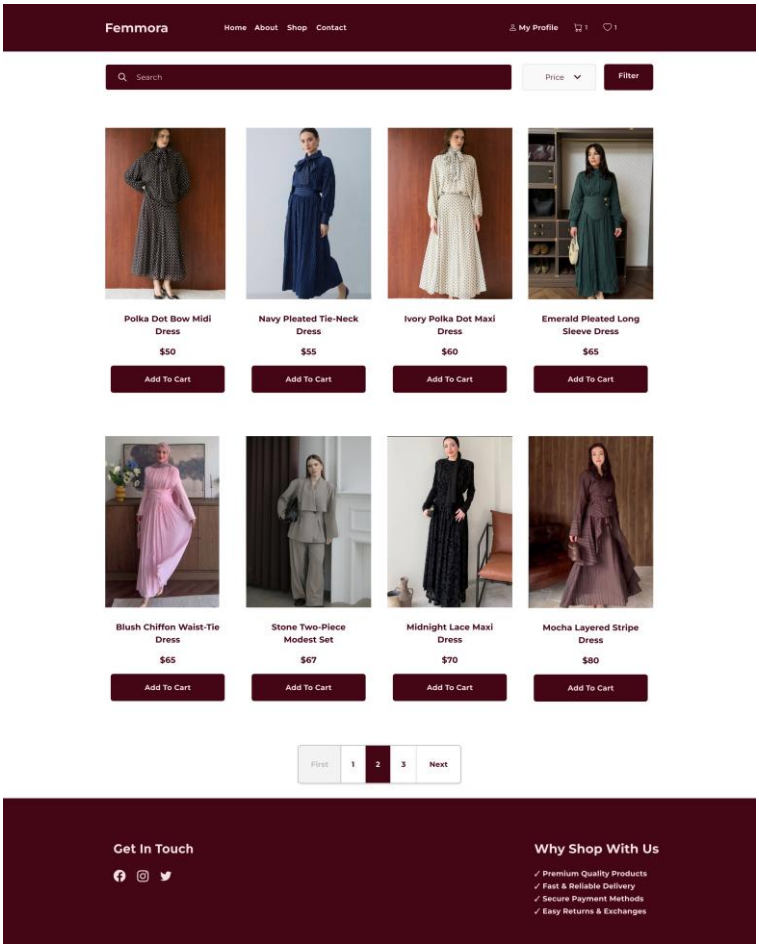
Order #	Name	Size	Price	Status	Date
#12345	Navy Pleated Tie-Neck Dress	M	$55	Preparing	1/6/2026
#12346	Polka Dot Bow Midi Dress	S	$50	Delivering	14/2/2026
#12347	Emerald Pleated Long Sleeve Dress	M	65$	Delivered	1/8/2025
#12348	1. Ivory Polka Dot Maxi Dress 2. Navy Pleated Tie-Neck Dress	1. M 2. S	$115	Delivered	30/3/2024

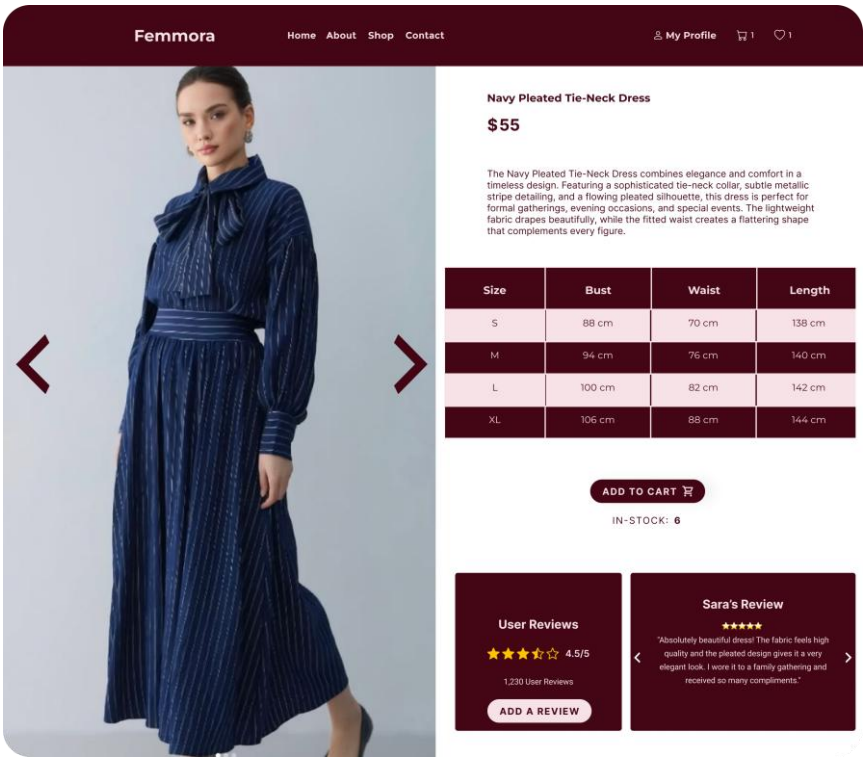


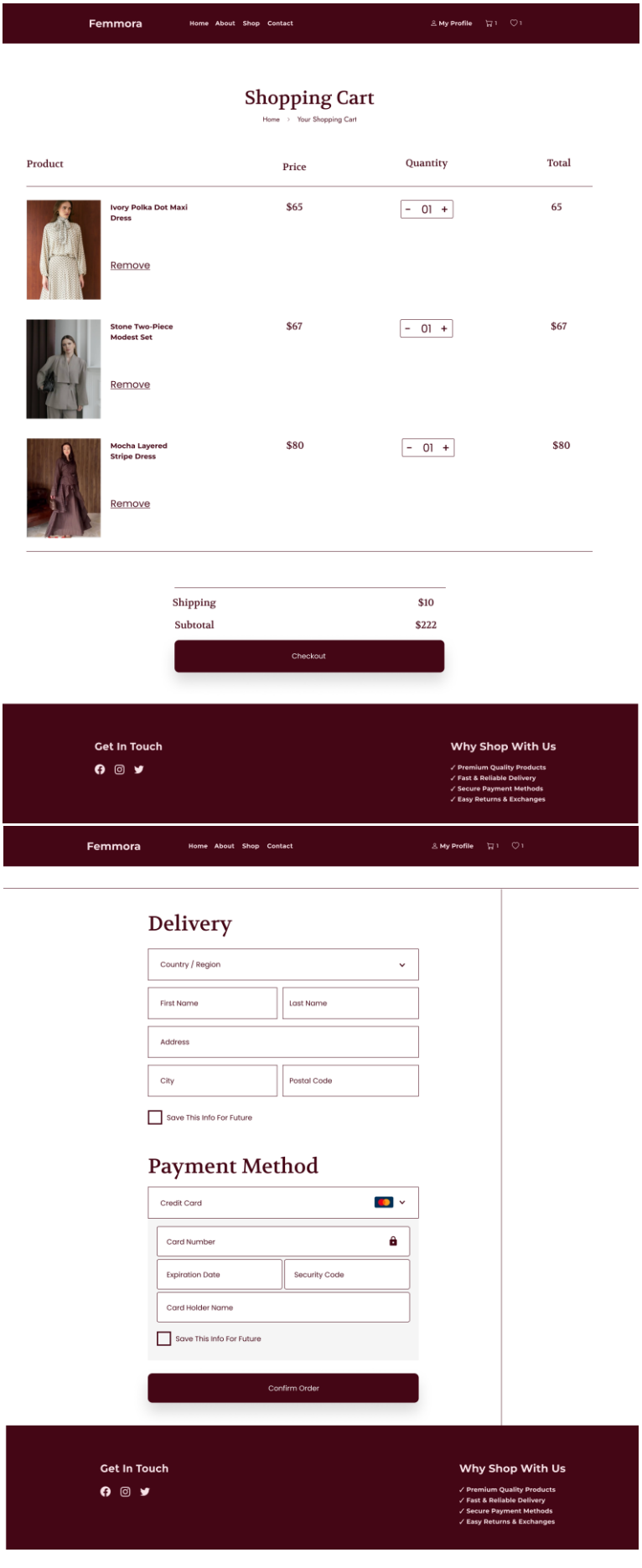


Payment Method DropDown Menu

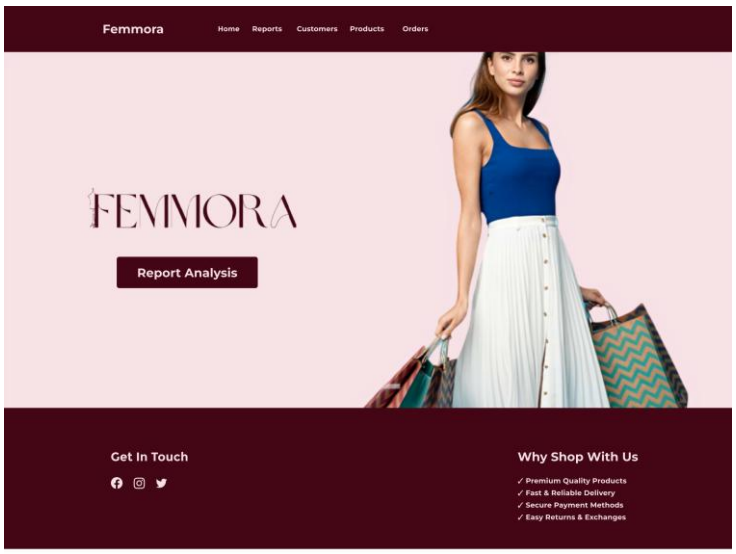


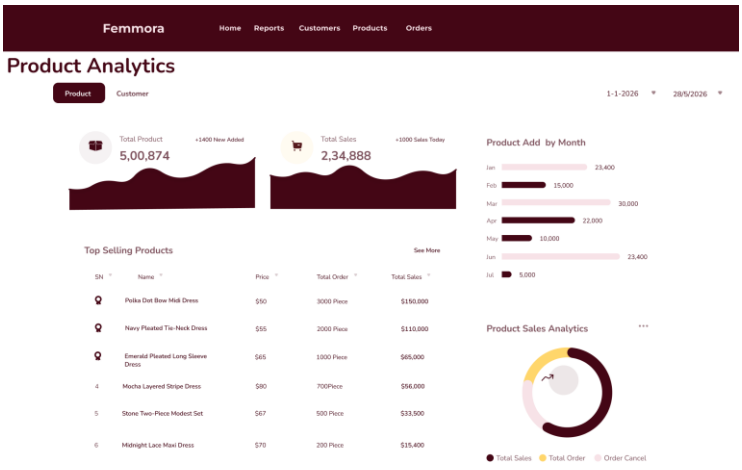

Femmora
Home About Shop Contact
My Profile
Search
Price
Filter
Polka Dot Bow Midi Dress
$50
Add To Cart
Navy Pleated Tie-Neck Dress
$55
Add To Cart
Ivory Polka Dot Maxi Dress
$60
Add To Cart
Emerald Pleated Long Sleeve Dress
$65
Add To Cart
Blush Chiffon Waist-Tie Dress
$65
Add To Cart
Stone Two-Piece Modest Set
$67
Add To Cart
Midnight Lace Maxi Dress
$70
Add To Cart
Mocha Layered Stripe Dress
$80
Add To Cart
First
1
2
3
Next
Get In Touch
Why Shop With Us
✓ Premium Quality Products
✓ Fast & Reliable Delivery
✓ Secure Payment Methods
✓ Easy Returns & Exchanges

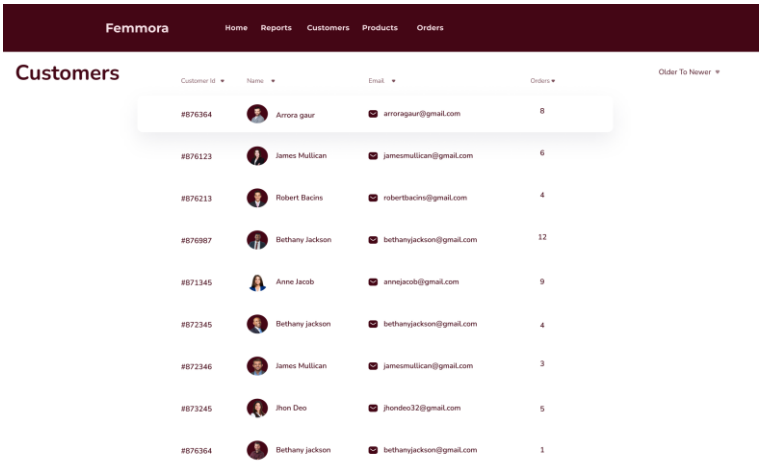

Femmora
Home About Shop Contact
My Profile
Navy Pleated Tie-Neck Dress
$55
The Navy Pleated Tie-Neck Dress combines elegance and comfort in a timeless design. Featuring a sophisticated tie-neck collar, subtle metallic stripe detailing, and a flowing pleated silhouette, this dress is perfect for formal gatherings, evening occasions, and special events. The lightweight fabric drapes beautifully, while the fitted waist creates a flattering shape that complements every figure.
Size
Bust
Waist
Length
S
88 cm
70 cm
138 cm
M
94 cm
76 cm
140 cm
L
100 cm
82 cm
142 cm
XL
106 cm
88 cm
144 cm
ADD TO CART
IN-STOCK: 6
User Reviews
4.5/5
1,230 User Reviews
ADD A REVIEW
Sara's Review
"Absolutely beautiful dress! The fabric feels high quality and the pleated design gives it a very elegant look. I wore it to a family gathering and received so many compliments."

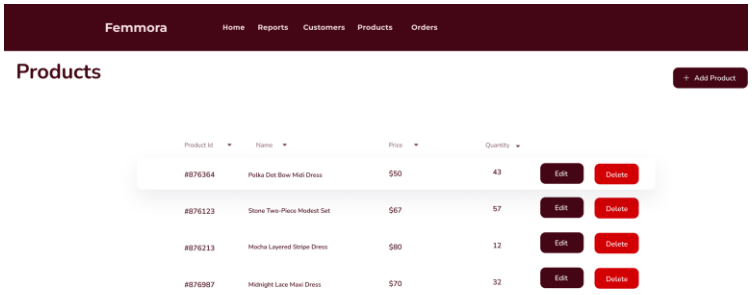

Femmora
Home About Shop Contact
My Profile
Shopping Cart
Home › Your Shopping Cart
Product
Price
Quantity
Total
Ivory Polka Dot Maxi Dress
$65
- 01 +
65
Remove
Stone Two-Piece Modest Set
$67
- 01 +
$67
Remove
Mocha Layered Stripe Dress
$80
- 01 +
$80
Remove
Shipping
$10
Subtotal
$222
Checkout
Get In Touch
Why Shop With Us
✓ Premium Quality Products
✓ Fast & Reliable Delivery
✓ Secure Payment Methods
✓ Easy Returns & Exchanges
Femmora
Home About Shop Contact
My Profile
Delivery
Country / Region
First Name
Last Name
Address
City
Postal Code
Save This Info For Future
Payment Method
Credit Card
Card Number
Expiration Date
Security Code
Card Holder Name
Save This Info For Future
Confirm Order
Get In Touch
Why Shop With Us
✓ Premium Quality Products
✓ Fast & Reliable Delivery
✓ Secure Payment Methods
✓ Easy Returns & Exchanges

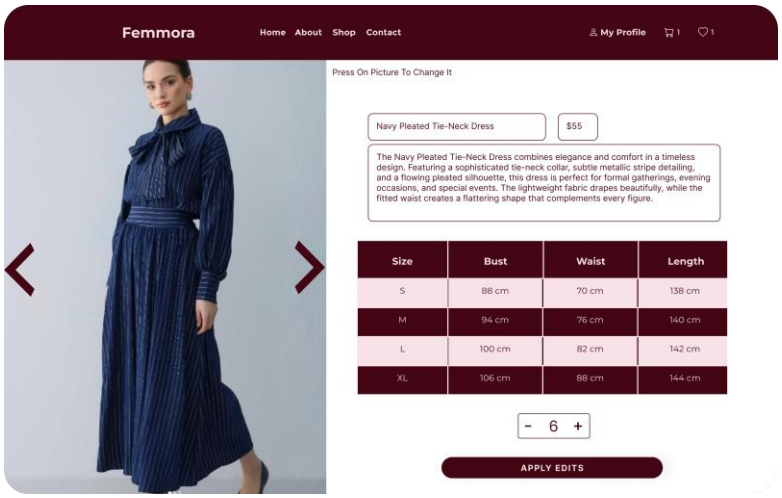
Admin

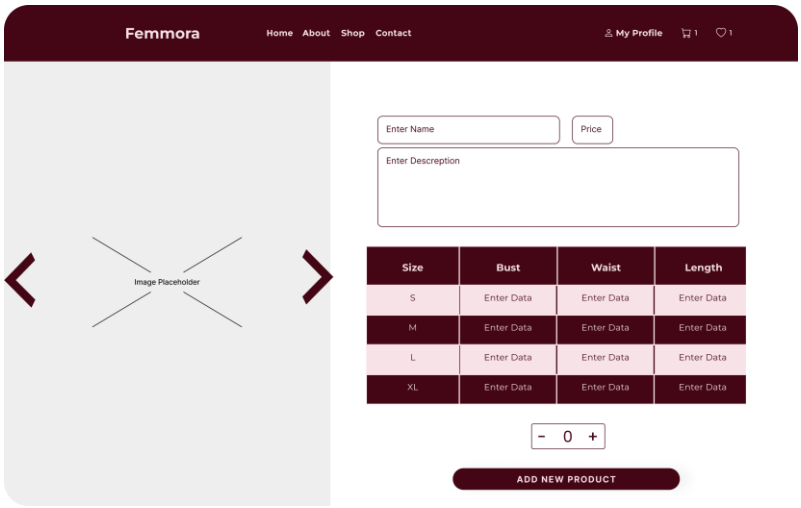

Femmora
Home
Reports
Customers
Products
Orders
FEMMORA
Report Analysis
Get In Touch
Why Shop With Us
✓ Premium Quality Products
✓ Fast & Reliable Delivery
✓ Secure Payment Methods
✓ Easy Returns & Exchanges

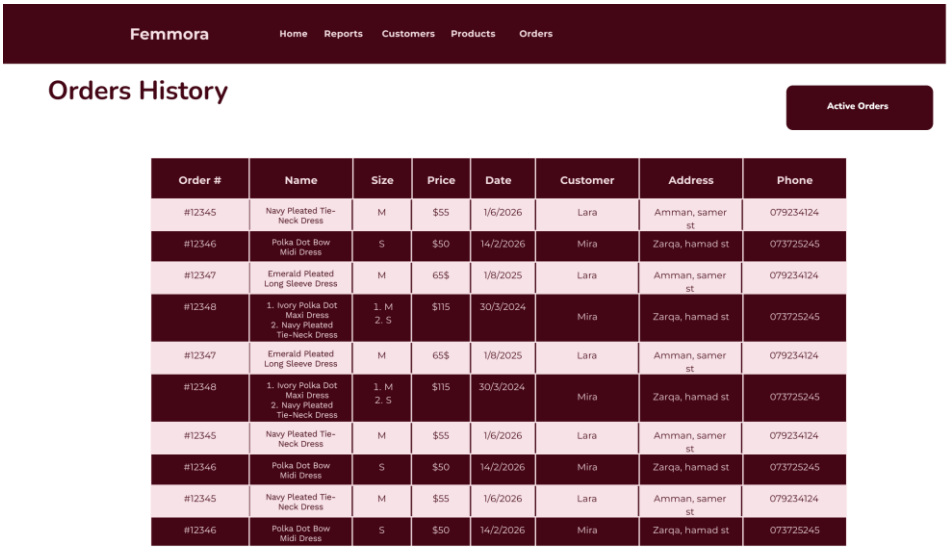

Femmora
Home
Reports
Customers
Products
Orders
Product Analytics
Product
Customer
1-1-2026
28/5/2026
Total Product
5,00,874
Total Sales
2,34,888
Product Add by Month
Top Selling Products
See More
Product Sales Analytics
Total Sales
Total Order
Order Cancel


Femmora
Home
Reports
Customers
Products
Orders
Customers
Older To Newer
#B76364 Arrora gaur arroragaur@gmail.com 8
#B76123 James Mullican jamesmullican@gmail.com 6
#B76213 Robert Bacins robertbacins@gmail.com 4
#B76987 Bethany Jackson bethanyjackson@gmail.com 12
#B71345 Anne Jacob annejacob@gmail.com 9
#B72345 Bethany jackson bethanyjackson@gmail.com 4
#B72346 James Mullican jamesmullican@gmail.com 3
#B73245 Jhon Deo jhondeo32@gmail.com 5
#B76364 Bethany jackson bethanyjackson@gmail.com 1


Femmora
Home
Reports
Customers
Products
Orders
Products
+ Add Product
Product Id
Name
Price
Quantity
#876364
Polka Dot Bow Midi Dress
$50
43
Edit
Delete
#876123
Stone Two-Piece Modest Set
$67
57
Edit
Delete
#876213
Mocha Layered Stripe Dress
$80
12
Edit
Delete
#876987
Midnight Lace Maxi Dress
$70
32
Edit
Delete


Femmora
Home
About
Shop
Contact
My Profile
1
1
Press On Picture To Change It
Navy Pleated Tie-Neck Dress
$55
The Navy Pleated Tie-Neck Dress combines elegance and comfort in a timeless design. Featuring a sophisticated tie-neck collar, subtle metallic stripe detailing, and a flowing pleated silhouette, this dress is perfect for formal gatherings, evening occasions, and special events. The lightweight fabric drapes beautifully, while the fitted waist creates a flattering shape that complements every figure.
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Bust
Waist
Length
S
88 cm
70 cm
138 cm
M
94 cm
76 cm
140 cm
L
100 cm
82 cm
142 cm
XL
106 cm
88 cm
144 cm
- 6 +
APPLY EDITS


Femmora
Home
About
Shop
Contact
My Profile
1
1
Image Placeholder
Enter Name
Price
Enter Descreption
Size
Bust
Waist
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ADD NEW PRODUCT



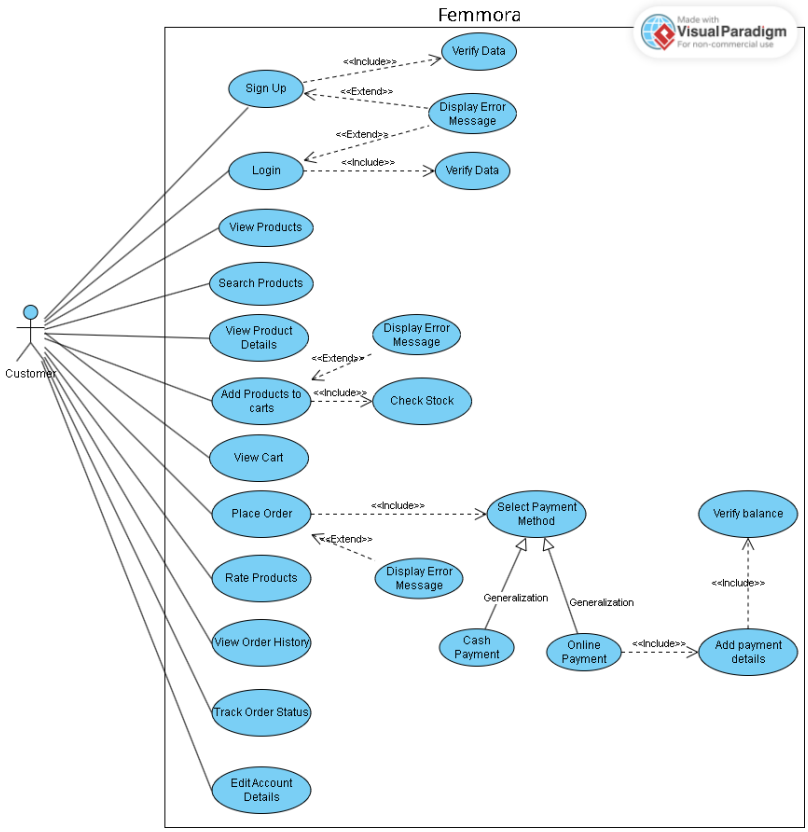
Order #	Name	Size	Price	Date	Customer	Address	Phone
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Lara	Amman, samer st	079234124
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Mira	Zarqa, hamad st	073725245
#12347	Emerald Pleated Long Sleeve Dress	M	65$	1/8/2025	Lara	Amman, samer st	079234124
#12348	1. Ivory Polka Dot Maxi Dress 2. Navy Pleated Tie-Neck Dress	1. M 2. S	$115	30/3/2024	Mira	Zarqa, hamad st	073725245
#12347	Emerald Pleated Long Sleeve Dress	M	65$	1/8/2025	Lara	Amman, samer st	079234124
#12348	1. Ivory Polka Dot Maxi Dress 2. Navy Pleated Tie-Neck Dress	1. M 2. S	$115	30/3/2024	Mira	Zarqa, hamad st	073725245
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Lara	Amman, samer st	079234124
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Mira	Zarqa, hamad st	073725245
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Lara	Amman, samer st	079234124
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Mira	Zarqa, hamad st	073725245



Active Orders

Order #	Name	Size	Price	Date	Status	Customer	Address	Phone
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Pick an option	Lamar	Ajloun, hamed St	07328145324
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Preparing	Sara	Amman, Sharhabil St	0794324521
#12347	Emerald Pleated Long Sleeve Dress	M	65$	1/8/2025	Pick an option	Lamar	Ajloun, hamed St	07328145324
#12348	1. Ivory Polka Dot Maxi Dress 2. Navy Pleated Tie-Neck Dress	1. M 2. S	$115	30/3/2024	Deleviring	Sara	Amman, Sharhabil St	0794324521
#12347	Emerald Pleated Long Sleeve Dress	M	65$	1/8/2025	Pick an option	Lamar	Ajloun, hamed St	07328145324
#12348	1. Ivory Polka Dot Maxi Dress 2. Navy Pleated Tie-Neck Dress	1. M 2. S	$115	30/3/2024	Pick an option	Sara	Amman, Sharhabil St	0794324521
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Preparing	Lamar	Ajloun, hamed St	07328145324
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Pick an option	Sara	Amman, Sharhabil St	0794324521
#12345	Navy Pleated Tie-Neck Dress	M	$55	1/6/2026	Pick an option	Lamar	Ajloun, hamed St	07328145324
#12346	Polka Dot Bow Midi Dress	S	$50	14/2/2026	Pick an option	Sara	Amman, Sharhabil St	0794324521

3.2 Use Case Diagrams



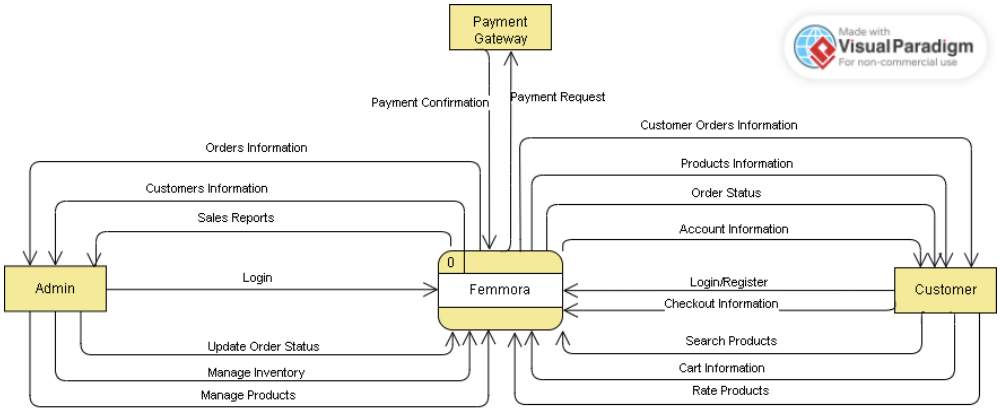


The use case diagram is a behavioural diagram that shows all the interactions that the user can do within the Femmora System. Include relationships were used when a use

case must always be executed as part of another use case such as verification of user data during registration and login. Extend relationships were used for optional or exceptional scenarios that do not always happen such as displaying an error message when invalid data is entered. Generalization was used to represent different payment methods, as cash payment and online payment are forms of payment selection process.

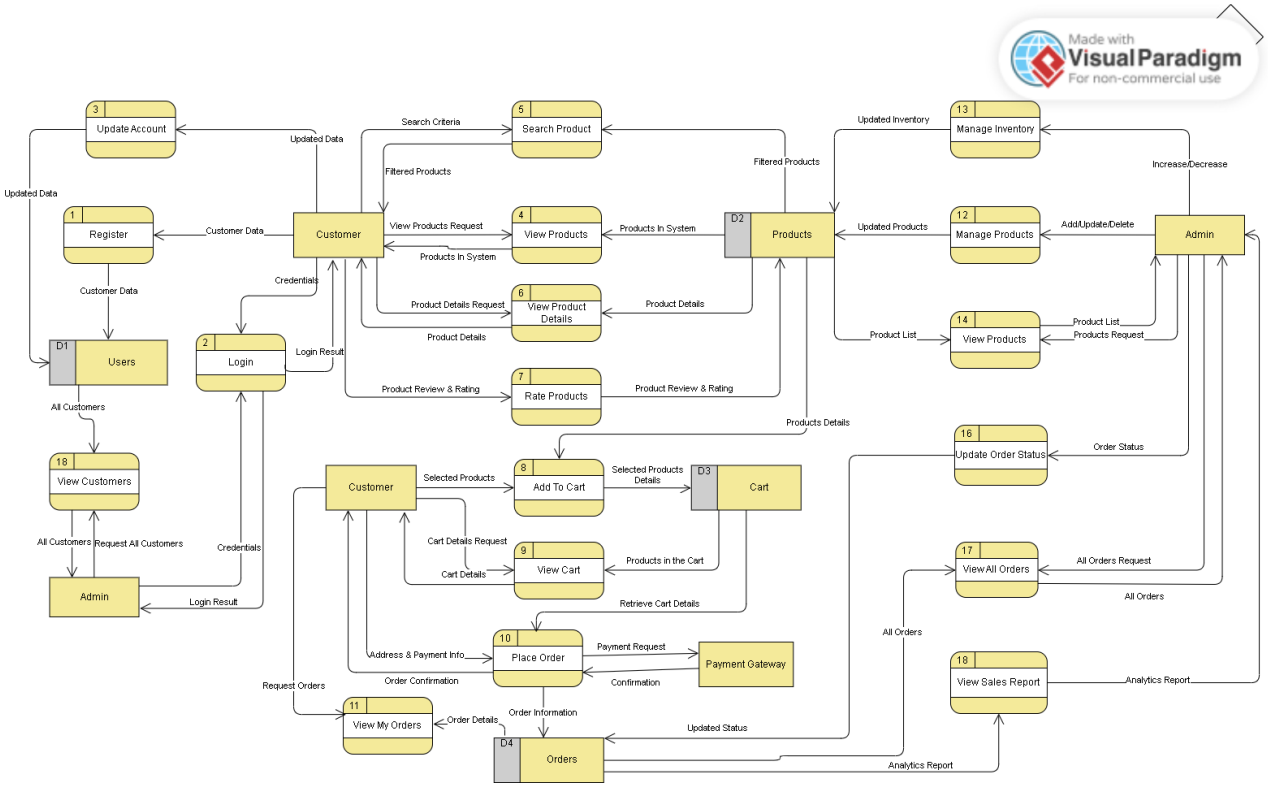
3.3 Data Flow Diagrams

Context Level



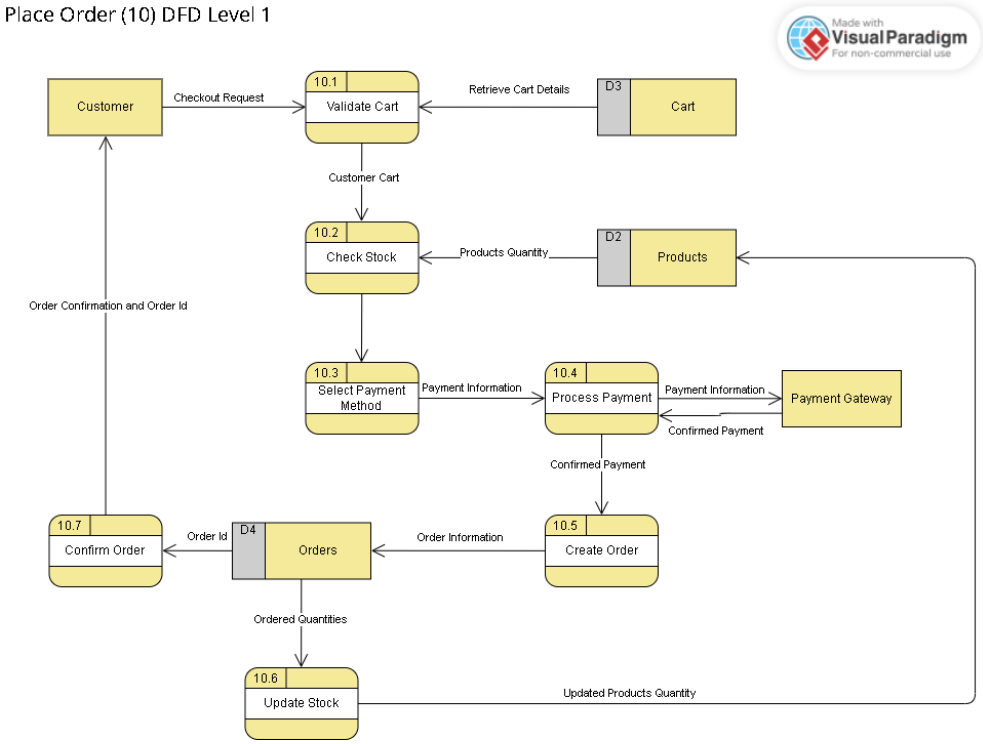
The Context Level of DFD represents the Femmora system as a single process and shows how it interacts with external entities such as customers, admin, and payment gateway. The diagram identifies the external entities and the information exchanged between them.

Level 0



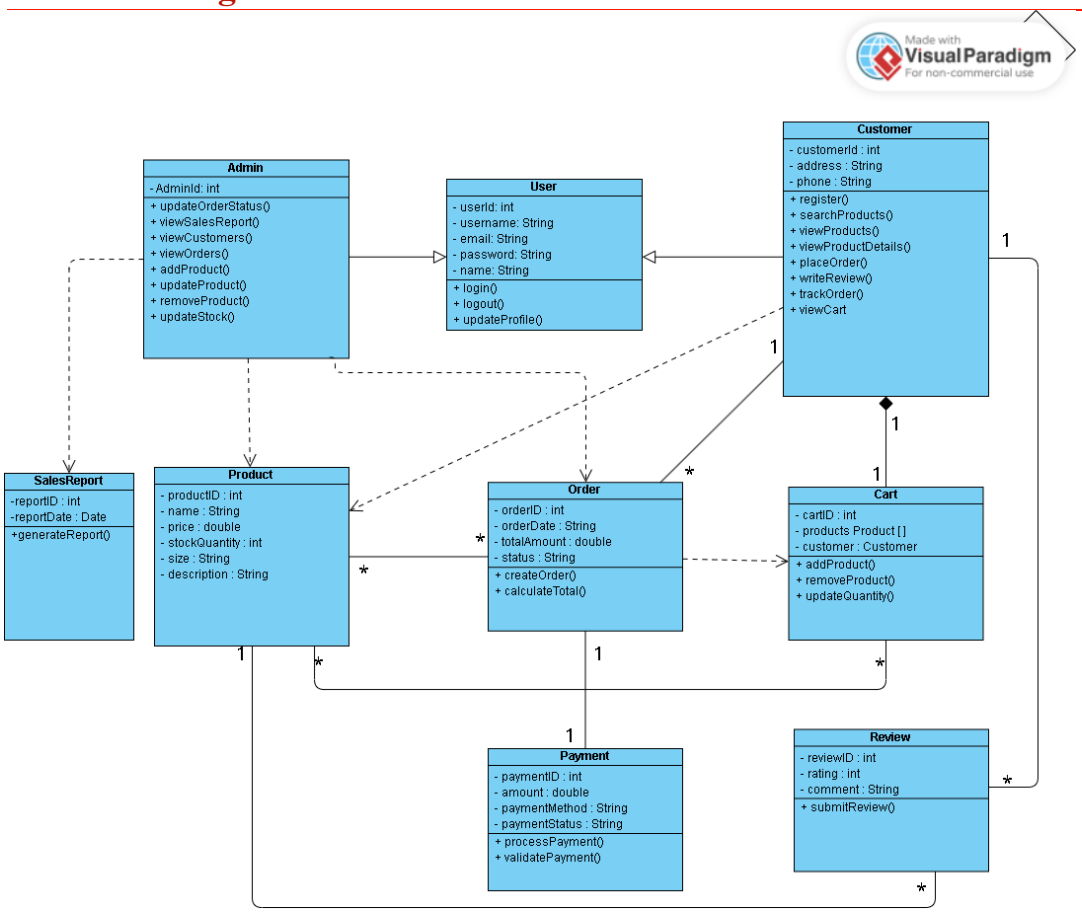
The level 0 of the DFD show a more detailed view of the Femmora system by including its processes and show how data flow between these processes, external entities, and data stores. Data stores such as Orders, Users, and Products show where information is stored and retrieved. There are 2 Admin, and 2 Customer entities to maintain the readability of the diagram.

Level 1



The level 1 of the DFD expands the processes into multiple sub-processes, and in my case, I expanded the place order process because it is the most complex process in the system. Each sub-processes represent the actual steps when customer want to place a new order, while the data flows show how information moves between customer, cart data store, products data store, payment gateway, and order data store. The choice of not making a validation of the payment because it happens internally in the payment gateway logic not in the Femmora system logic.

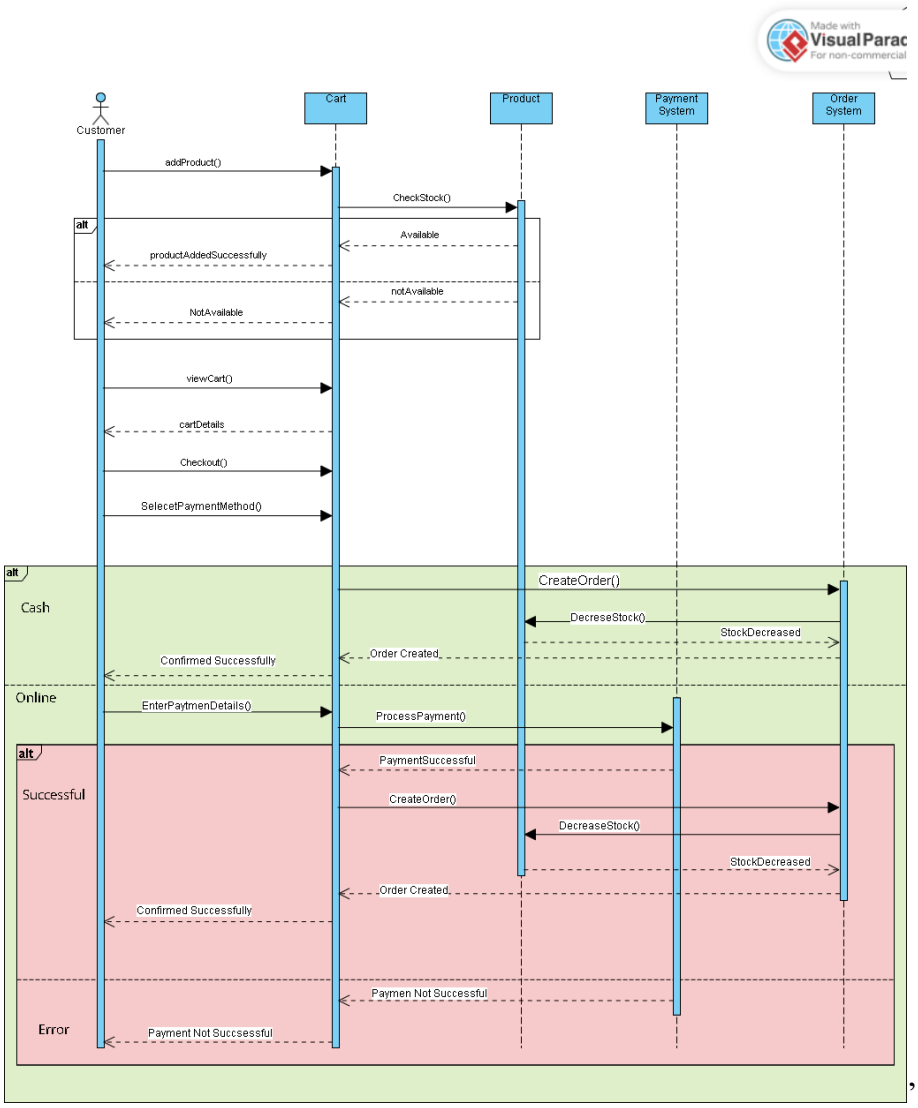
3.4 Class Diagrams



The class diagram gives a detailed structural view of the system by showing the system classes, their attributes, operations, and the relationships between these classes. The generalization between user and both customer and admin is because both actors share same user information, but they have different responsibilities. The composition between cart and customers show that each cart belongs to a one customer and it cannot exist without a customer. Dependency used between classes because one class depends on the other to do a specific operation such as admin update product. Association relationships were used:

- Customer - Order: A customer can place multiple orders, while each order belongs to one customer.
- Customer - Review: A customer can write multiple review, while each review belongs to one customer.
- Review - Product: A product can have multiple reviews, while each review belongs to one product.
- Product - Cart: A cart can contain many products, and a product can appear in multiple carts.
- Product - Order: An order can have multiple products, and a product can appear in multiple orders.
- Order - Payment: Only one order can have only one payment.

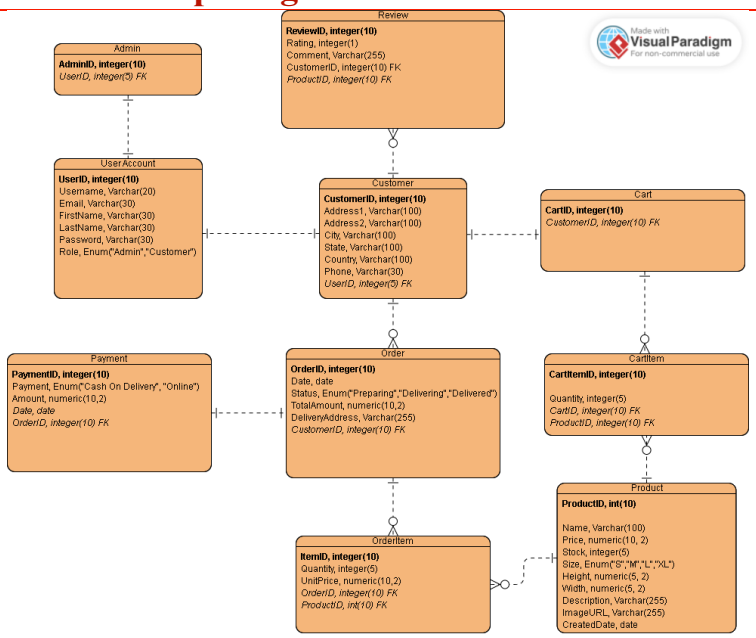
3.5 Sequence Diagrams



The sequence diagram shows the objects and the messages between them over time, like it focuses on the interactions and how these objects communicate.

The sequence diagram started when a customer adds a product to the cart, the system verifies the availability of the product using the alternative box. Then customer can view cart then process checkout and select payment method. There are alternative box inside an alternative box because when the customer chooses to pay in cash the system does not need to check the payment details but when the customer chooses to pay online the system must check the payment details including the balance and if its correct, so after checking the payment, the system creates the order, decreases stock quantity, and confirm the order, but if there was an error in the payment, the system does not create the order and inform the customer that payment failed.

3.6 Entity Relationship Diagrams



ERD shows the database structure of Femmora's system and it contains different entities, their attributes, and the relationships between those entities.

Relationshops were designed based on the business needs, as customer can place multiple orders, while each order belongs to one customer. Products and orders have many-to-many relationships which resolved through OrderItem entity. Also, CartItem entity was introduced because of the many-to-many relationships between Cart and Product. One to one relationship describe only one of that side and also only one for the other side, for example, one customer can have only one user account or one cart.

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